DANIEL SILVA

Junior Computer Engineer · AI & Data Analysis
Santa Maria da Feira, Aveiro, Portugal | 915 031 868 | danielsilva.eng.pt@gmail.com | LinkedIn | GitHub

PROFILE

I finished my degree in Computer Engineering at ISEC, in the Information Systems track, and have completed 3 curricular internships, including one in Spain through Erasmus+. I also won the Medal of Excellence at WorldSkills Portugal 2023 (Mobile Application Development, national level). Along the way I realised I want to work in data and AI, as a Data Analyst, Data/AI Engineer or Backend Python Developer. I like learning by doing, and I want to keep growing and deepening my knowledge.

PROFESSIONAL EXPERIENCE

Software Development — Curricular Internship — OPTIMA – Equipamentos CNC *Apr 2023 – Jun 2023 · Aveiro*

- I built from scratch a Python program to control a 3D modelling machine, with complex trigonometry and Euclidean geometry calculations.
- I built the graphical interface (GUI) and connected the program in real time to the CNC controller, reading log and NCP files.
- I tested everything directly on the machine, to make sure it worked in real conditions.

Full-Stack Web Developer — Erasmus+ Internship — Internet Web Solutions *Apr 2023 – May 2023 · Málaga, Spain*

- I built the "WaterWise" application on my own, to calculate water footprint, using PHP and a MySQL database.
- I used AJAX and jQuery to process data in real time and make the interface more dynamic.
- I also handled the responsive layout and some CSS3 animations, with a focus on user experience.

IT & Hardware — Curricular Internship — OPTIMA – Equipamentos CNC *Jun 2022 – Jul 2022 · Aveiro*

- I set up, calibrated and operated 3D printers and scanners (Calibry), and modelled technical parts in Fusion 360.
- I gave technical support in an industrial environment, including calibrating laser cutting machines at client sites (LaserGRBL).
- This is where I started working with automation in Python, dealing with the logic of Cartesian axes (XYZ).

RELEVANT PROJECTS

- **IndustrIA** (personal project): a question-answering system over industrial maintenance manuals, built on a RAG (Retrieval-Augmented Generation) architecture — PDF extraction and chunking, embeddings (all-MiniLM-L6-v2) stored in PostgreSQL with pgvector, cosine similarity search and answer generation via an LLM (Groq), exposed through a FastAPI API.
- **BI system for the Port of Figueira da Foz** (university group project): a Data Mart in a Snowflake schema, with automated ETL in Python (combining MySQL data and CSV files) and Power BI dashboards to track operational KPIs.
- **Skin disease classifier with Deep Learning** (university group project): I used Transfer Learning with MobileNetV2 on a Kaggle dataset, tuned with Random Search, and reached 72% accuracy, compared against a baseline model. The app is available online via Streamlit (dermoscanai.streamlit.app).
- **WaterWise Calculator:** the water footprint application mentioned above, built with PHP, MySQL and AJAX.
- **Python CNC Modelling Controller:** the automation software for a 3D modelling machine that I built during my internship at OPTIMA.

TECHNICAL SKILLS

Languages	Python, SQL, PL/SQL, Java, C++, C
Data & AI	Data Analysis, ETL, Machine Learning / Deep Learning (Transfer Learning), Power BI, Data Warehousing (Snowflake Schema)
Databases	PostgreSQL, MySQL, SQL Server
Tools	Git/GitHub, Streamlit

EDUCATION

BSc in Computer Engineering — Information Systems track *2023 – 2026*
Instituto Superior de Engenharia de Coimbra (ISEC)

Technical Diploma in IT Systems Management and Programming *2020 – 2023*
Escola Secundária Dr. Serafim Leite

AWARDS

- Medal of Excellence — WorldSkills Portugal, 2023 (Mobile Application Development, national level)